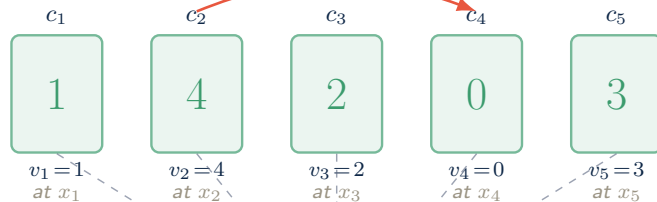


deck = covering curve $C = \text{Bring's curve} \subset \mathbb{P}^4$

Equations: $\sum_i x_i = 0, \sum_i x_i^2 = 0, \sum_i x_i^3 = 0$ (S_5 permutes the five coordinates)

$g \in S_5$: permutes the coordinates x_i



reconstruction

compute $\sum_i v_i \bmod 5$;
any $T=5$ cards give s .
fewer than 5:
 s stays uniform over $\{0, \dots, 4\}$.

**[hidden]
secret
 s**

$$\begin{aligned} s &= \sum_i v_i \bmod 5 \\ &= (1+4+2+0+3) \bmod 5 \\ &= 10 \bmod 5 \\ &= 0 \end{aligned}$$

base curve $\mathbb{P}^1$

$t \in \mathbb{P}^1$ parameterises $s \in \{0, 1, 2, 3, 4\}$; s stays hidden until $\geq T=5$ cards combine

geometry

$$\begin{aligned} \pi : (x_1 : \dots : x_5) &\mapsto t \\ t &= e_5(x)^2 / e_4(x)^5 \end{aligned}$$

$$\begin{aligned} e_4 &= \sum_{i < j < k < l} x_i x_j x_k x_l \\ e_5 &= x_1 x_2 x_3 x_4 x_5 \end{aligned}$$

protocol

$s = (v_1 + v_2 + v_3 + v_4 + v_5) \bmod 5$
both are S_5 -symmetric:
 s depends only on the orbit,
i.e. on $t = \pi(\text{card})$